

Comparison with EMA

The currently accepted flood-frequency methodology in the United States, described in Bulletin 17C (U.S. Geological Survey, 2018), recommends fitting the LPIII distribution using the Expected Moments Algorithm (EMA). EMA is a major improvement over the Bulletin 17B guidelines (U.S. Geological Survey, 1982), and is capable of incorporating historical and paleoflood information into flood frequency studies.

EMA was developed as an alternative to MLE and the Bulletin 17B (U.S. Geological Survey, 1982) method for incorporating historical information in flood frequency studies (Cohn, Lane, & Baier, 1997). EMA was shown to achieve greater efficiency than the B17B adjustment for censored data, and nearly achieving the efficiency of MLE while avoiding some of the numerical complications. However, in the cases where there is no censored data, EMA and B17B are identical and exhibit substantial bias as compared to MLE (Cohn, Lane, & Baier, 1997).

It would be unreasonable to expect a Bayesian estimation approach to perfectly match a moments-based fitting approach, such as EMA, or vice versa. The two estimation methods are very different from a theoretical perspective. Nevertheless, comparisons of RMC-BestFit and EMA were performed to provide some insight into how the methods may differ in practice. These comparisons do not validate or invalidate either method.

RMC-BestFit was compared with EMA for the 82 USGS gage sites (see Table 1 for a listing) used for testing in Bulletin 17C. Several additional sites have been compared during risk assessments performed as part of the USACE Dam Safety program, including studies with historical and paleoflood data. In general, for most sites, RMC-BestFit produces similar results to EMA. In cases where there is only systematic data, EMA is identical with the method of moments (MOM), which means the estimators can be biased and less efficient than MLE and Bayesian estimation. Consequently, in those cases, the quantile standard error and confidence intervals for EMA are sometimes wider than RMC-BestFit. In addition, MOM can be more sensitive than Bayesian estimation for sites with only one or two events that are much larger than the rest of the data. In these scenarios, the data likelihood used for MLE and Bayesian estimation is less influenced by single large values.

An example of this behavior is seen at site 01439500 Bush Kill at Shoemaker, PA, where out of 102 years of record, the largest event is more than double the next highest peak. There is only systematic data in this example. As can be seen in Figure 60, the EMA fit pulls upwards towards the highest peak as compared to the RMC-BestFit results. Furthermore, the EMA confidence intervals are considerably wider than the credible intervals from RMC-BestFit. A comparison of the computed parameters and resulting log-likelihood is provided in Table 92. As expected, the posterior mode from RMC-BestFit produces a slightly higher log-likelihood than the EMA fit.

Table 92 – Comparison of RMC-BestFit and EMA parameters for USGS 01439500 Bush Kill at Shoemaker, PA.

Parameter	EMA	RMC-BestFit
Mean (of log) (μ)	3.3327	3.3327
Std. Dev (of log) (σ)	0.2322	0.2306
Skew (of log) (γ)	0.9725	0.7839
Log-Likelihood	-858.3656	-857.4655

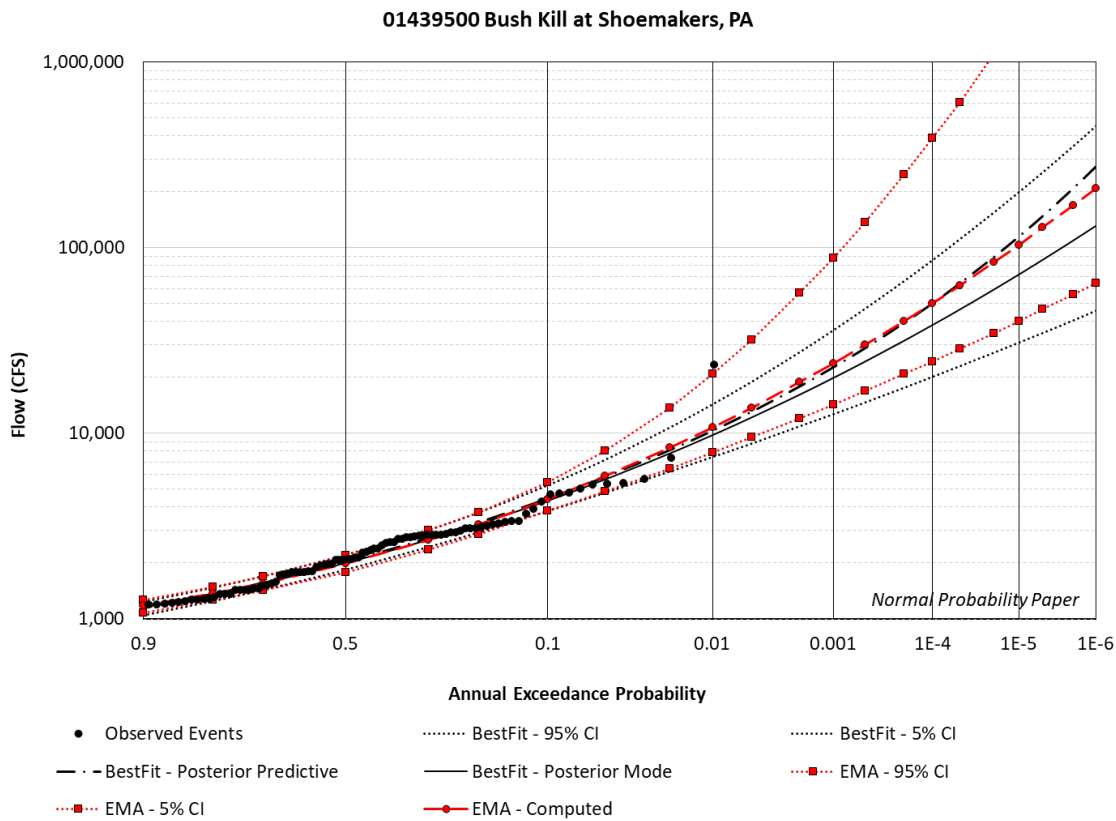


Figure 60 – Comparison of RMC-BestFit and EMA for USGS 01439500 Bush Kill at Shoemaker, PA.

Poor agreement between RMC-BestFit and EMA can also be expected in cases where there are many low outliers and the data is very negatively skewed. If the absolute value of the skewness coefficient is greater than 2, then the MLE method cannot produce a solution for the shape parameter of the PIII or LPIII (Bobee & Ashkar, 1991). This implies that the MLE estimate for the shape parameter is biased. If the absolute value of skew is greater than approximately 1.25, it is not recommended to use PIII/LPIII with MLE (Bobee & Ashkar, 1991). If the absolute value of skew is greater than 2, the method of “conditional maximum likelihood” must be applied. Conditional MLE is performed by first fixing the location parameter to the smallest value of the sample if the skew is positive, or the largest value of the sample if the skew is negative. Then, the scale and shape parameters are solved for, conditionally on the fixed location parameter. In cases where the absolute value of skew is greater than 1.25 and approaching 2, Bayesian estimation will automatically produce results that are similar to conditional MLE.

Example #2 from Appendix 10 of Bulletin 17C (U.S. Geological Survey, 2018) provides a good example of this behavior. This example is from USGS site 11274500 Orestimba Creek near Newman, CA, which has 30 low outliers. The results from RMC-BestFit for this site are shown in Figure 61. The skewness of the data is approximately -1.75. The posterior mode for the location parameter of LPIII is ~15,000, which is very close the largest value of the sample of 12,000 cfs. This demonstrates that the Bayesian approach will automatically produce results similar to conditional MLE. A comparison of RMC-BestFit with EMA is provided in Figure 62. Both methods produce similar confidence intervals, but the computed curve from EMA provides a less biased fit for the right-hand tail. When integrating over the confidence intervals from EMA, an expected probability curve can be computed. Figure 63 shows that the posterior predictive distribution from RMC-BestFit closely matches the expected probability curve from EMA for this site.

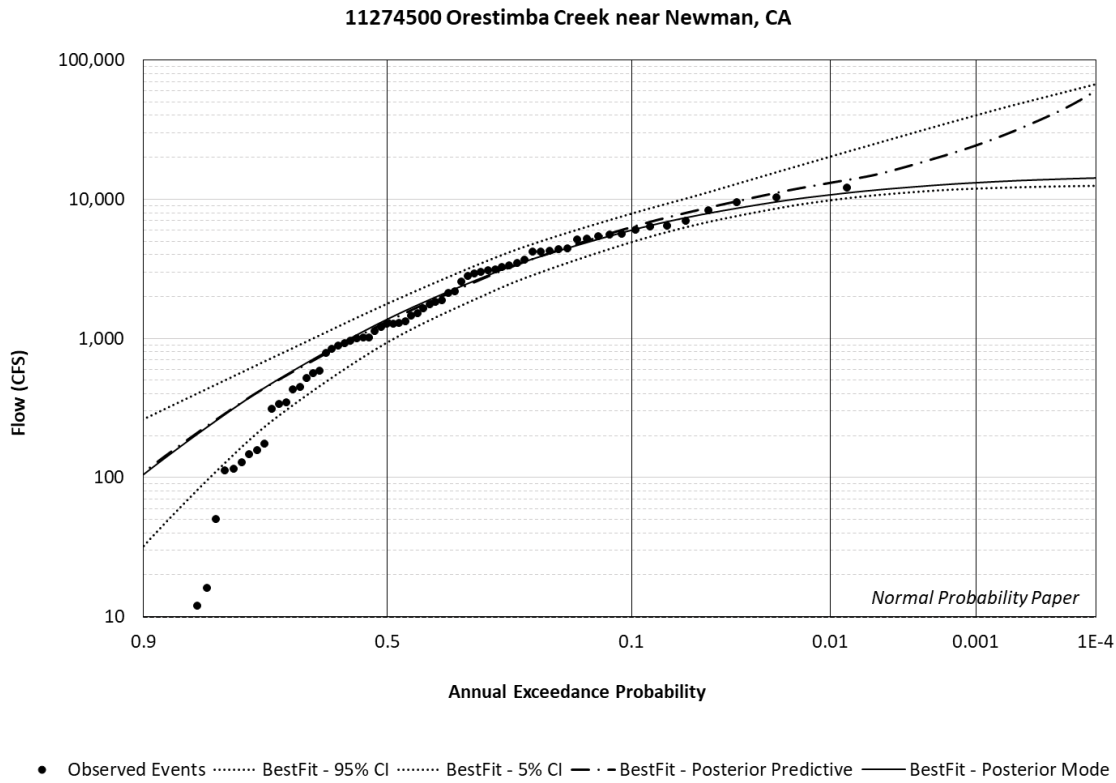


Figure 61 – RMC-BestFit Results for USGS 11274500 Orestimba Creek near Newman, CA.

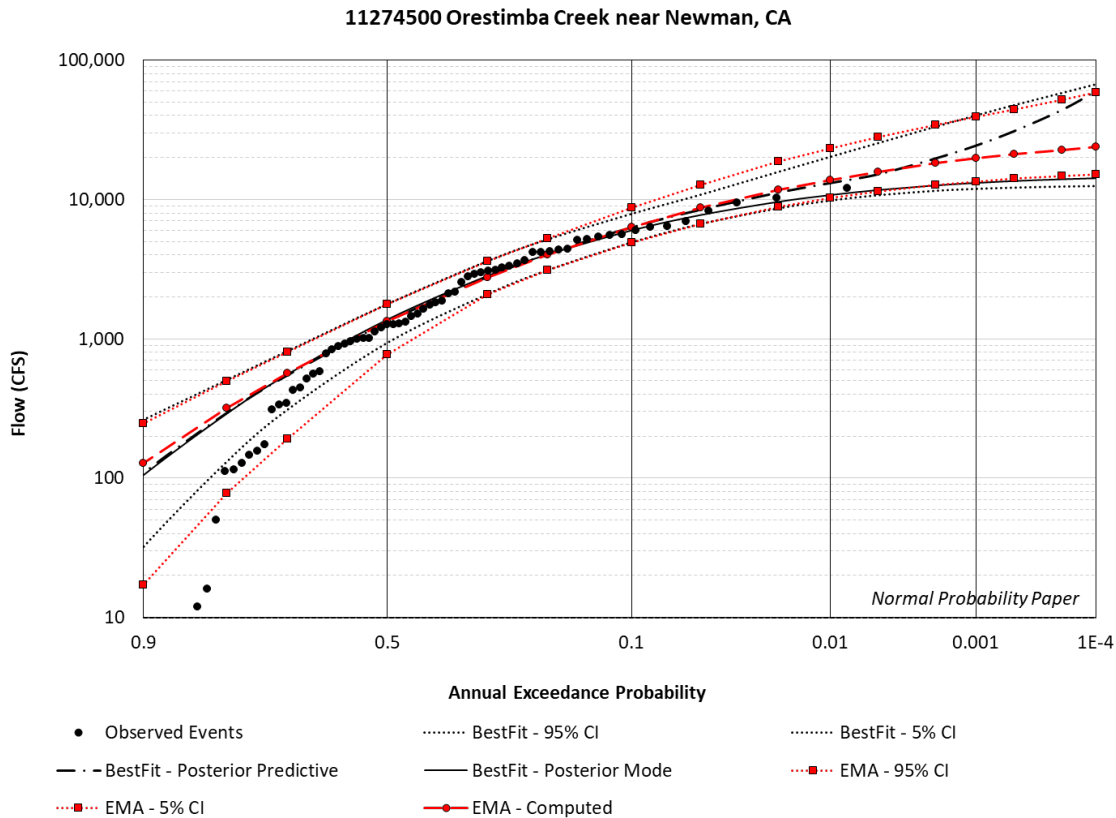


Figure 62 – Comparison of RMC-BestFit and EMA for USGS 11274500 Orestimba Creek near Newman, CA.

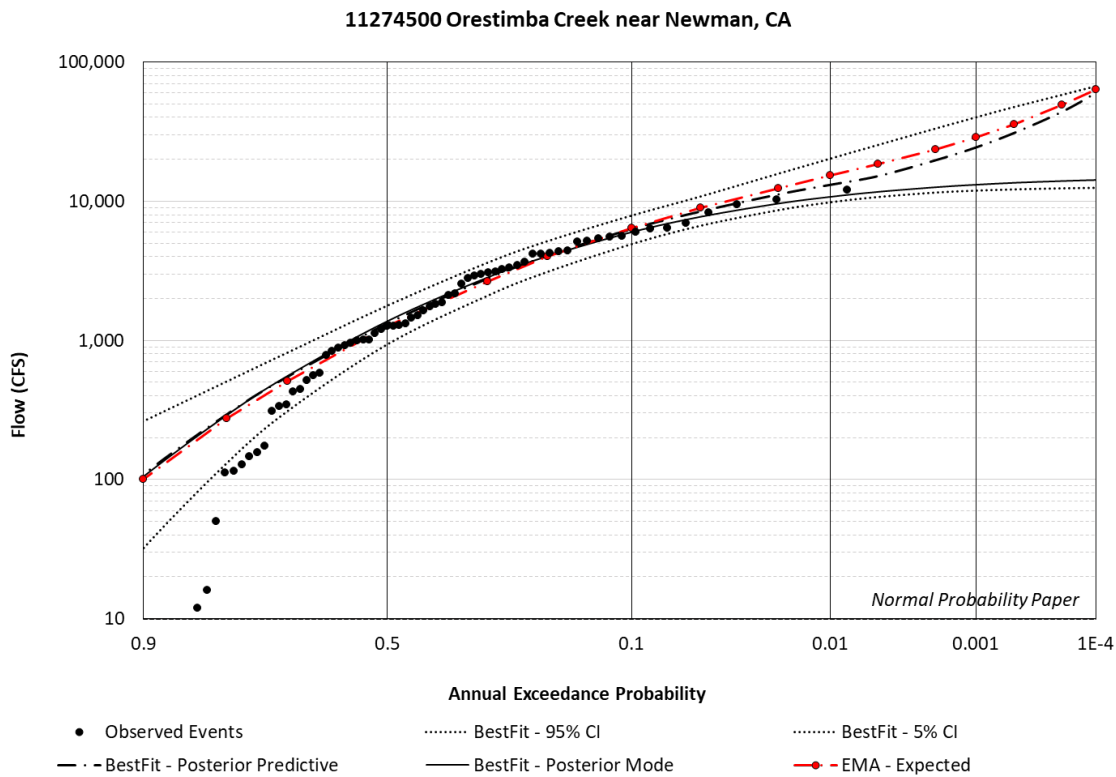


Figure 63 – Comparison of RMC-BestFit Posterior Predictive with the EMA Expected Probability Curve for USGS 11274500 Orestimba Creek near Newman, CA.

In cases where there is historical data incorporated with intervals and perception thresholds, RMC-BestFit and EMA will generally produce similar results. For example, RMC-BestFit was compared to EMA for site 11446500 American River at Fair Oaks, CA, which is example #7 from Appendix 10 in Bulletin 17C (U.S. Geological Survey, 2018). In this example, there is a major historical flood in 1862 and four paleoflood events dating back over 1,000 years. Results of the comparison are shown in Figure 64. For this example, the differences between EMA and RMC-BestFit are inconsequential, and the computed curve from EMA and the posterior mode from RMC-BestFit, along with the confidence intervals, are very consistent.

A final comparison was performed using Example #10 provided in the *HEC-SSP* example download. This example is for USGS station 01470500 Schuylkill River, PA, and it includes historical data, a perception threshold and regional skew. The regional skew is 0.001 with an MSE of 0.064. In RMC-BestFit, the regional skew is entered as an informative prior on skew that is Normally distributed with a mean of 0.001 and standard deviation 0.253. As shown in Figure 65, RMC-BestFit and EMA produce virtually identical results for this example.

A more comprehensive investigation into the similarities and differences between RMC-BestFit and EMA, and the respective limitations of each, is planned for a future report.

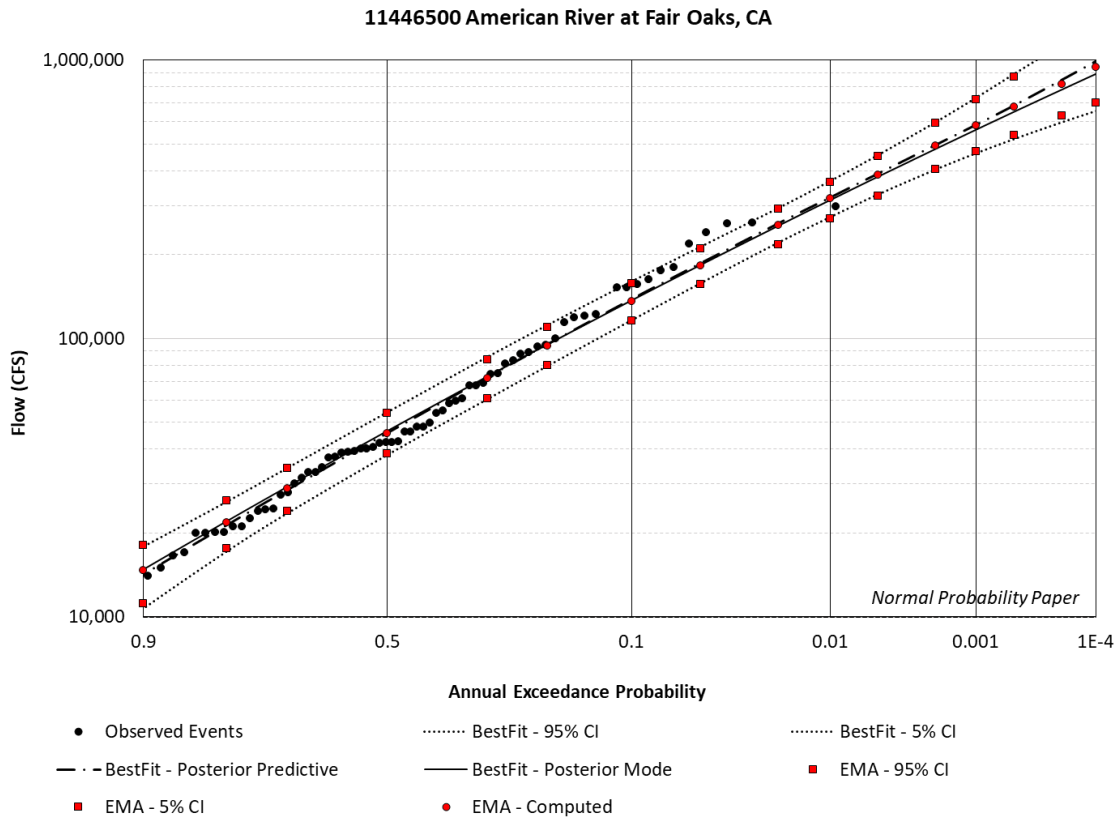


Figure 64 – Comparison of RMC-BestFit and EMA for USGS 11446500 American River at Fair Oaks, CA.

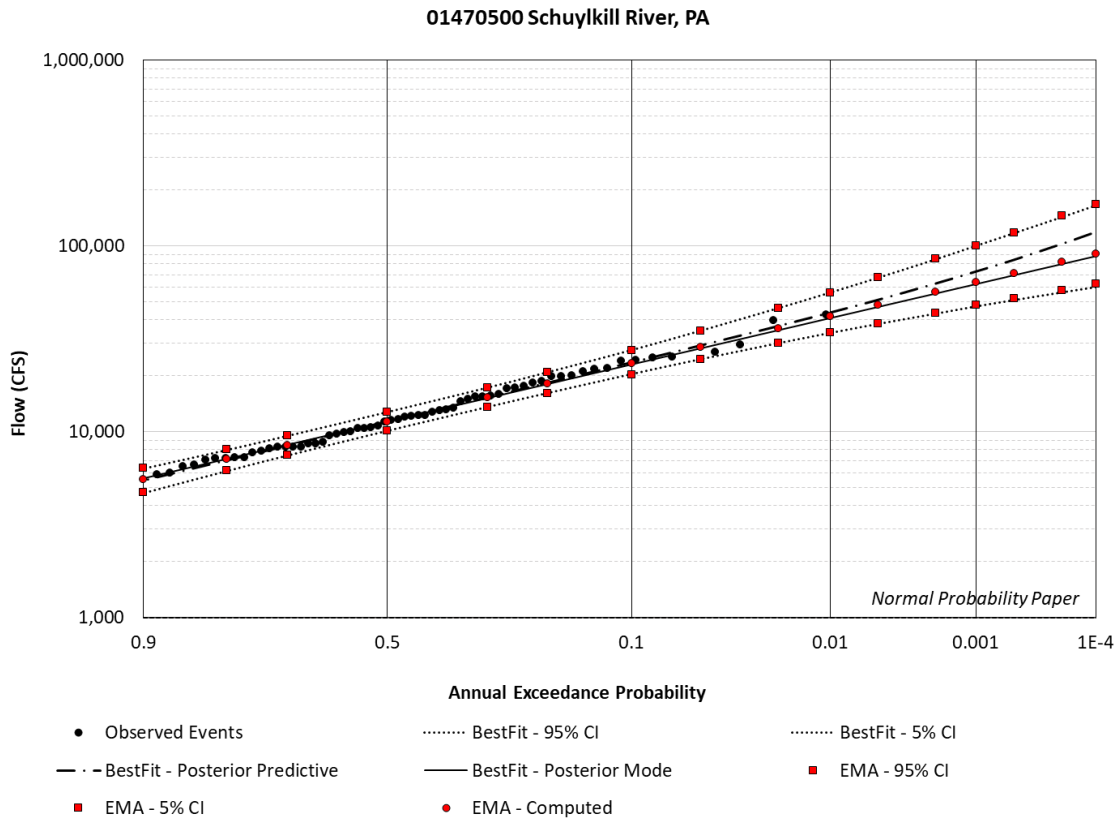


Figure 65 – Comparison of RMC-BestFit and EMA for USGS 01470500 Schuylkill River, PA.